

Appendix A: Nominal

SEAGLIDER - TRIM AND BALANCE

SG 194

Date

24-Oct-13

Vehicle CG

x_CG

95.752

cm

z_CG

-0.835

cm

y_CG

0

cm

Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m
(8x) 10-32 x 1/2 FHMS, joint ring to aft fairing	55279	1	14.3	0.0143	116.8	0.210	0.00	0.00835	0.000	0
(8x) 10-32 x 1/2 FHMS, Ogive Aft fairing to endcap	55279	1	14.3	0.0143	125.7	0.299	0.00	0.00835	0.000	0
Forward Fairing - with foam	49897	1	4237.1	4.2371	71.7	-0.241	0.00	0.00835	0.000	0
Joint Ring, Fwd to Aft Fairing (bonded to fwd fairing before paint; Long ring = 875g, short ring 688g)	49834	1	688	0.688	116.5	0.207	0.00	0.00835	0.000	0
Nose weight (bonded to fwd fairing after paint)	49844	1	1270.9	1.2709	9.4	-0.864	0.00	0.00835	0.000	0
Nose weight plates	55448	1	495.3	0.4953	9.4	-0.864	0.00	0.00835	0.000	0
(2x) 1/4-20 x 1.5° SHCS + 1/4 LW brass, nose plate to nose base (2 -3 plates)	55185/55253	1	23.4	0.0234	9.4	-0.864	0.00	0.00835	0.000	0
Trim Lead .600" aft of pupa joint ring - bottom (5" strip)		1	695.1	0.6951	106.9	0.111	-13.10	-0.12265	0.000	0
EPDM base (5"x6") + (2x) 2"x 6" tape - bottom		1	24.2	0.0242	106.9	0.111	-12.70	-0.11865	0.000	0
Trim Lead .600" aft of pupa joint ring - port side (5" strip)		1	759.8	0.7598	106.9	0.111	0.00	0.00835	0.000	0
EPDM base (5"x6") + (2x) 2"x 6" tape - port		1	24.0	0.024	106.9	0.111	0.00	0.00835	0.000	0
Trim Lead 600" aft of pupa joint ring - starboard side (5" strip)		1	865.0	0.865	106.9	0.111	0.00	0.00835	0.000	0
EPDM base (5"x6") + (2x) 2"x 6" tape - starboard		1	24.0	0.024	106.9	0.111	0.00	0.00835	0.000	0
Trim Lead tape (~1g per 5" lead strip)		15	12.0	0.012	106.9	0.111	13.00	0.13835	0.000	0
(2) Straps - cut to length (start at 34g, subtract trimmed)		1	33.2	0.0332	106.9	0.111	0.00	0.00835	0.000	0
(2) wraps 2x around black elec tape for security		1	12.0	0.012	106.9	0.111	0.00	0.00835	0.000	0
Ogive Aft Fairing - with cutouts for panels	56972	1	2535	2.535	143.2	0.474	-0.10	0.00735	0.000	0
Ogive Wing, Starboard (1m)	49838	1	617.8	0.6178	129.779	0.340	0.00	0.00835	29.905	0.29905
Ogive Wing, Port (1 m)	49838	1	611.3	0.6113	129.779	0.340	0.00	0.00835	-29.905	-0.29905
(8x) 10-32 x 1/2 FHMS, wings to fairing	5xxxx	1	14.3	0.0143	129.6	0.338	0.00	0.00835	0.000	0
Top Rudder			62.7	0.0627	170.784	0.750	10.91	0.10079	0.000	0
Bottom Rudder		1	62.7	0.0627	170.784	0.750	-10.91	-0.10079	0.000	0
(2x) 1/4-20 x 1 1/2" FHMS, rudder to rudder shoe	5xxxx	1	15.7	0.0157	181.8	0.860	0.00	0.00835	0.000	0
Ogive Top panel (with sensor holes)	57651	1	298.1	0.2981	142.8	0.470	11.80	0.12635	0.000	0
Bottom panel with sensor holes	57651	1	290.4	0.2904	142.8	0.470	-11.80	-0.10965	0.000	0
(20x) Ogive Panel screws - 6-32 x 3/8 Truss head MS	56912	1	19.2	0.0192	142.8	0.470	0.00	0.00835	0.000	0
Top panel tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)	NIB	4	14.8	0.0148	132.1	0.363	13.30	0.14135	0.000	0
Bottom panel tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		4	14.8	0.0148	143	0.472	13.30	0.14135	0.000	0
Aft-end strb Tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		2	7.4	0.0074	161.3	0.655	0.00	0.00835	0.000	0
Aft-end port Tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		2	7.4	0.0074	161.3	0.655	0.00	0.00835	0.000	0
Under strb-wing Tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		4	14.8	0.0148	132.1	0.363	0.00	0.00835	0.000	0
Under port-wing Tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		4	14.8	0.0148	132.1	0.363	0.00	0.00835	0.000	0
Trim lead to Port & Starbord of puck		1	526	0.526	154.9	0.591	-10.20	-0.09365	0.000	0
Trim lead Aft of puck		1	260.2	0.2602	167.6	0.718	-10.20	-0.09365	0.000	0
Tape for trim lead on bottom panel		3	2.4	0.0024	157.5	0.617	-10.20	-0.09365	0.000	0

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cm

y_CG

0

cm

Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m
Tubes & glue mass on top hatch (4) - non-compressee	NIB	1	71.5	0.0715	154.9	0.591	-13.10	-0.12265	0.000	0
Compressee only on top hatch (4)		1	877.7	0.8777	154.9	0.591	-13.10	-0.12265	0.000	0
Tubes & glue mass on bottom hatch (2)		1	36	0.036	154.9	0.591	-12.70	-0.11865	0.000	0
Compressee only on bottom hatch (2)		1	403.1	0.4031	154.9	0.591	-12.70	-0.11865	0.000	0
Tubes & glue mass under port wing (4)		1	71.4	0.0714	132.1	0.363	0.00	0.00835	0.000	0
Compressee only under port wing (4)		1	876.6	0.8766	132.1	0.363	0.00	0.00835	0.000	0
Tubes & glue mass under starboard wing (4)		1	71.2	0.0712	132.1	0.363	0.00	0.00835	0.000	0
Compressee only under starboard wing (4)		1	896.6	0.8966	132.1	0.363	0.00	0.00835	0.000	0
Tubes & glue mass aft on port side (2)		1	35.6	0.0356	161.3	0.655	0.00	0.00835	0.000	0
Compressee only aft on port side (2)		1	385.8	0.3858	161.3	0.655	0.00	0.00835	0.000	0
Tubes aft & glue mass aft on starboard side (2)		1	35.6	0.0356	161.3	0.655	0.00	0.00835	0.000	0
Compressee only aft on starboard side (2)		1	427.1	0.4271	161.3	0.655	0.00	0.00835	0.000	0
Antenna Assembly GPSI (All parts), including extra comms cable	52336 / 52341 / 52454 / 52476 / 55361	1	628.7	0.6287	211.84	1.161	0.00	0.00835	0.000	0
O2 Sensor (Aanderas optode)	52220 / 52485 / 52486 / 55239 / 55321	1	293.0	0.293	142.9	0.471	15.90	0.16735	0.000	0
O2 Sensor Assy. Hardware (mount base, SS hose clamp, (2x) 8-32 x 5/8 FHMS, (2x) 8-32 nylok)	52220 / 52485 / 52486 / 55239 / 55321	1	34.7	0.0347	142.9	0.471	15.90	0.16735	0.000	0
Wet Labs Sensor		1	234.7	0.2347	154.9	0.591	-10.20	-0.09365	0.000	0
Wet Labs sensor Assy. Hardware ((2x) snesor clamp, 2(x) 8-32 x 5/8 FHMS - sensor clamp to fairing)	55381 / 52312 / 55245	1	3.2	0.0032	154.9	0.591	-10.20	-0.09365	0.000	0
Position D (SBE-O2): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-2.90	-0.02065	0.000	0
Position E (WL-2): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-5.10	-0.04265	0.000	0
Position F (WL-3 / PAR): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-5.80	-0.04965	0.000	0
Position G (?): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-5.10	-0.04265	0.000	0
Position H (?): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-2.90	-0.02065	0.000	0
Position I (?): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	2.90	0.03735	0.000	0
O-Ring E70-270, Endcap to Batt Hull	52411	1	8.0	0.008	123.3	0.275	0.00	0.00835	0.000	0
(2x) 6-32 x 1/2 SHCS w/ LW + (5x) 6-32 x 5/8 with zincs, endcap to battery hull	55183 / 55217 / 55214 / 49884	1	13.2	0.0132	123.5	0.277	0.00	0.00835	0.000	0
O-Ring, E70-270, Bulkhead to batt. hull	52411	1	8.0	0.008	72.9	-0.229	0.00	0.00835	0.000	0
(2x) 6-32 x 1/2 SHCS w/ LW + (5x) 6-32 x 5/8 with zincs, bulkhead to battery hull	55183 / 55217 / 55214 / 49884	1	15.3	0.0153	65.4	-0.304	0.00	0.00835	0.000	0
(4x) 6-32 x 1/2 SHCS + #6 LW, rails to bulkhead	55183 / 55217	1	5.3	0.0053	59.5	-0.363	0.00	0.00835	0.000	0
O-Ring, E70-263, electric hull to bulkhead	52410	1	6.4	0.0064	66.4	-0.294	0.00	0.00835	0.000	0
(8x) 6-32 x 1/2 SHCS + \$6 LW, elec hull to bulkhead	55183 / 55217	1	10.5	0.0105	65.8	-0.300	0.00	0.00835	0.000	0
(8x) Ogive fairing adapter clips		1	43.7	0.0437	65.8	-0.300	0.00	0.00835	0.000	0

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(8x) 1/4-28 x 3/8 PHMS, Ogive fairing adapter clip to aft endcap		1	37.1	0.0371	65.8	-0.300	0.00	0.00835	0.000	0
Endcap, Aft	49815	1	2027.6	2.0276	127.0	0.312	0.00	0.00835	0.000	0
(2x) Roll pin, 5/32 OD x 3/8, Pump anti rotation	55248	1	1.0	0.001	123.0	0.272	0.00	0.00835	0.000	0
Connector, Run Tee, 1/8 NPT x 1/4 tube, Push to connect		1	8.4	0.0084	125.2	0.294	0.00	0.00835	0.000	0
Connector, Male, 7/16 straight x 1/4 tube, SS	55506	1	27.6	0.0276	125.2	0.294	1.30	0.02135	0.000	0
Assy, Solenoid Valve, Skinner	52254	1	600.7	0.6007	124.5	0.287	1.50	0.02335	0.000	0
Adaptor, Male, 1/8 NPT x 1/4 M, SS	52430	1	10.5	0.0105	128.1	0.323	1.30	0.02135	0.000	0
Connector, Male, 1/4 tube x 1/8 NPT, Push to connect		1	6.3	0.0063	123.9	0.281	-1.30	-0.00465	0.000	0
(3x) Tubing, 1/4 DIA x 10"		1	12.1	0.0121	119.4	0.236	0.00	0.00835	0.000	0
(2x) Tubing, 1/4 Dia x 4.375"		1	3.5	0.0035	119.4	0.236	0.00	0.00835	0.000	0
Tubing, 1/4 Dia x 1.25"		1	0.5	0.0005	119.4	0.236	0.00	0.00835	0.000	0
Tubing, 1/4 DIA x 8"		1	3.2	0.0032	119.4	0.236	0.00	0.00835	0.000	0
Check Valve, Kepner 204 A-1	52448	1	23.0	0.023	124.7	0.289	-1.30	-0.00465	0.000	0
Branch Tee, Male, 1/4 tube x 1/8 NPT, Push to connect		1	8.4	0.0084	125.2	0.294	0.00	0.00835	0.000	0
Seal Washer	52350	1	0.9	0.0009	129.3	0.335	0.00	0.00835	0.000	0
LeDuc Output Screw	52349	1	7.8	0.0078	129.6	0.338	0.00	0.00835	0.000	0
Hydraulic Pump, LeDuc	52348	1	317.2	0.3172	129.6	0.338	0.00	0.00835	0.000	0
Hydraulic endcap	49803	1	111.1	0.1111	125.9	0.301	0.00	0.00835	0.000	0
Ball Bearing, 8 mm ID x 16 mm OD	52459	1	3.1	0.0031	121.2	0.254	0.00	0.00835	0.000	0
O-Ring, N70-032, hydraulic endcap	52407	1	0.5	0.0005	121.1	0.253	0.00	0.00835	0.000	0
Connector, Male, 1/16 NPT x 1/8 tube, Push to connect		1	5.9	0.0059	122.5	0.267	2.90	0.03735	0.000	0
Tubing, 1/8 Dia. x 10.5"	52418	1	0.9	0.0009	121.3	0.255	0.00	0.00835	0.000	0
Magnetic Seal	52353	1	13.9	0.0139	114.7	0.189	0.00	0.00835	0.000	0
(6x) 6-32 x 1/2 SHCS + #6 LW, hydraulic encap to encap	55183 / 55217	1	7.8	0.0078	121.1	0.253	0.00	0.00835	0.000	0
Lee Plug Set	52445 / 52446	1	0.4	0.0004	122.2	0.264	-1.30	-0.00465	0.000	0
Check Valve (altered item Kepner 1106 A-1-1) + O-Ring (N70 3-906)	49852 / 52408	1	48.1	0.0481	130.8	0.350	0.00	0.00835	0.000	0
Bladder, External	52441	1	837.8	0.8378	133.3	0.375	0.00	0.00835	0.000	0
Nut, Bladder	49802	1	107.3	0.1073	138.7	0.429	0.00	0.00835	0.000	0
Position A (Comms): IE55 w o-ring	52160	1	11.7	0.0117	131.1	0.353	5.10	0.05935	0.000	0
Position B (Opt O2): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	2.90	0.03735	0.000	0
Position C (WL-1): IE55 w o-ring	52160	1	11.5	0.0115	130.8	0.350	0.00	0.00835	0.000	0
Position D (SBE O2): IE55 w o-ring	52160	1	11.8	0.0118	130.8	0.350	-2.90	-0.02065	0.000	0
Position E (WL-2): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-5.10	-0.04265	0.000	0
Position F (WL-3): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-5.80	-0.04965	0.000	0
Position G (?): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-5.10	-0.04265	0.000	0
Position H (?): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-2.90	-0.02065	0.000	0
Position I (?): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	2.90	0.03735	0.000	0

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Position J (?): IE55 w o-ring	52160	1	11.8	0.0118	130.8	0.350	5.10	0.05935	0.000	0
Iridium/GPS Ant., conn/o-ring/ CA W5, J2	52241	1	66.1	0.0661	130.8	0.350	0.00	0.00835	0.000	0
Vaccum/pressure relief valve (deep sea power)	52449	1	11.5	0.0115	130.8	0.350	5.80	0.06635	0.000	0
Pump Plate, Inner	49804	1	37.8	0.0378	120.1	0.243	1.90	0.02735	0.000	0
Assy, Main pump motor w/ CA W8, W18	52253	1	130.7	0.1307	123.2	0.274	4.60	0.05435	0.000	0
(3x) M2 x 6 FHMS (motor to interface plate)	55203	1	0.4	0.0004	120.1	0.243	4.60	0.05435	0.000	0
Bore Reducer	52351	1	0.7	0.0007	118.8	0.230	0.00	0.00835	0.000	0
Pulley, LeDuc pump motor	52352	1	5.6	0.0056	118.8	0.230	4.70	0.05535	0.000	0
(6x) 4-40 x 3/8 FHMS, hyd end cap to inner plate	55229	1	1.9	0.0019	120.1	0.243	0.00	0.00835	0.000	0
Pulley, LeDuc pump	49806	1	91.7	0.0917	118.8	0.230	0.00	0.00835	0.000	0
Timing Belt	52357	1	3.4	0.0034	118.8	0.230	1.30	0.02135	0.000	0
Pump Plate, Outer w/ (4x) standoffs	49805	1	20.2	0.0202	118.1	0.223	1.90	0.02735	0.000	0
(4x) 4-40 x 3/16 FHCS		1	1.2	0.0012	118.4	0.226	0.00	0.00835	0.000	0
Check Valve, Kepner 404A-1		1	25.3	0.0253	125.2	0.294	0.00	0.00835	0.000	0
Connector, 45 elbow, 1/8 NPT x 1/4 tube, Push to connect		1	7.2	0.0072	125.2	0.294	0.00	0.00835	0.000	0
Connector, 90 elbow, 1/8 NPT x 1/4 tube, Push to connect		1	7.5	0.0075	125.2	0.294	0.00	0.00835	0.000	0
Connector, Union Tee, 1/4 tube, Push to connect		1	2.9	0.0029	125.2	0.294	0.00	0.00835	0.000	0
Pump Bracket		1	21.5	0.0215	124.2	0.284	0.00	0.00835	0.000	0
Motor, Chrg Pump w/ W34		1	59.9	0.0599	124.2	0.284	0.00	0.00835	0.000	0
(2x) M2 x 6 FHMS, motor to bracket		1	0.2	0.0002	124.2	0.284	2.50	0.03335	0.000	0
Magnet Rotor Housing		1	10.5	0.0105	124.2	0.284	0.00	0.00835	0.000	0
(4x) 6-32 x 1/4 PHMS, bracket to housing		1	1.8	0.0018	124.2	0.284	-1.00	-0.00165	0.000	0
Rotary Magnet		1	18.0	0.018	124.2	0.284	-1.00	-0.00165	0.000	0
Pumphead, Micropump + mounting plate + (3x) 4-40 x 3/8 FHMS		1	224.7	0.2247	124.2	0.284	-2.00	-0.01165	0.000	0
(4x) 4-40 x 3/16 SHCS + #4 LW, housing to flange		1	2.0	0.002	124.2	0.284	-1.30	-0.00465	0.000	0
(2x) Elbow, 1/8 not x 1/4 tube, push to connect		1	15.0	0.015	124.2	0.284	-2.30	-0.01465	0.000	0
Piston, Diaphragm	49811	1	135.3	0.1353	114.3	0.185	0.00	0.00835	0.000	0
Diaphragm, Internal (Bellofram)	52359	1	551.9	0.5519	115.7	0.199	0.00	0.00835	0.000	0
Sikaflex Adhesive Compound	55459	1	15.6	0.0156	114.3	0.185	0.00	0.00835	0.000	0
Cylinder Head, Diaphragm with extra port		1	336.4	0.3364	111.5	0.157	0.00	0.00835	0.000	0
Branch Tee, 1/4 tube x 1/8 NPT, push to connect; 1/4 tube plug		1	9.2	0.0092	109.8	0.140	-6.10	-0.05265	0.000	0
Elbow, 1/8 NPT M x 1/8 NPT M, brass	52426	1	26.9	0.0269	109.8	0.140	6.10	0.06935	0.000	0
Kepner Check Valve 404A-1-200		1	26.6	0.0266	109.8	0.140	2.00	0.02835	0.000	0
90 Elbow, 1/8 npt x 1/8 tube, push to connect		1	7.4	0.0074	109.8	0.140	-0.30	0.00535	0.000	0
Filter, Baston (25 micron) / with zip tie		1	20.9	0.0209	109.8	0.140	2.00	0.02835	0.000	0
Union, 1/4 tube, push to connect		1	1.8	0.0018	109.8	0.140	2.00	0.02835	0.000	0
90 Elbow, 1/8 npt x 1/4 tube, push to connect		1	7.5	0.0075	109.8	0.140	2.00	0.02835	0.000	0

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0

cm

Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m
Bleed screw, 10-32 x .125 PHMS with seal	55477	1	1.8	0.0018	109.8	0.140	7.60	0.08435	0.000	0
Cylinder, Diaphragm	49812	1	421.5	0.4215	115.1	0.193	0.00	0.00835	0.000	0
(8x) 5-40 x 3/8 SHCS + #5 LW, head to cylinder	55251	1	7.5	0.0075	110.6	0.148	0.00	0.00835	0.000	0
Standoff (starboard)	55299	1	12.2	0.0122	121.3	0.255	6.60	0.07435	0.000	0
Standoff (port)	55299	1	12.5	0.0125	121.3	0.255	-6.60	-0.05765	0.000	0
(4x) 6-32 x 3/8 SHCS, standoff to cylinder	55183	1	4.5	0.0045	120.7	0.249	0.00	0.00835	0.000	0
(2x) Saddles, mounting, Aft PCB + (4x) 4-40 x 1/4 PHMS	52373 / 55257	1	5.3	0.0053	115.6	0.198	8.10	0.08935	0.000	0
(2x) Saddles, mounting, C PCB + (4x) 4-40 x 1/4 PHMS	49888 / 55257	1	6.7	0.0067	115.6	0.198	3.80	0.04635	0.000	0
(2x) Saddles, mounting, T PCB + (4x) 4-40 x 1/4 PHMS	49889 / 55257	1	6.7	0.0067	115.6	0.198	-3.80	-0.02965	0.000	0
(2x) Saddles, mounting, SBE O2 PCB + (4x) 4-40 x 1/4 SHCS	49890 / 55257	1	4.5	0.0045	115.6	0.198	-8.10	-0.07265	0.000	0
Linear Potentiometer A w/ cable assy (W21), Linear pot clamp, 3-48 x 1/2 SHCS, Spring, Linear pot rod end	52255 / 49814 / 55220	1	40.9	0.0409	121.3	0.255	6.40	0.07235	0.000	0
Linear Potentiometer B w/ cable assy (W20), Linear pot clamp, 3-48 x 1/2 SHCS, Spring, Linear pot rod end	52256 / 49814 / 55220	1	38.5	0.0385	121.3	0.255	6.40	0.07235	0.000	0
(2x) 5-40 x 1/2 SHCS + #5 LW, Clamp to cylinder	55250 / 55212 / 55225	1	1.9	0.0019	121.0	0.252	0.00	0.00835	0.000	0
(2x) 6-32 x 1/2 SHCS + #6 LW, boost pump bracket to endcap	55183 / 55217	1	2.7	0.0027	122.6	0.268	0.00	0.00835	0.000	0
(4x) 6-32 x 1/2 SHCS + #6 LW, cylinder standoff to endcap	55184 / 55217	1	5.4	0.0054	121.7	0.259	0.00	0.00835	0.000	0
Aft Terminal PCB	55302	1	25.0	0.025	115.6	0.198	8.60	0.09435	0.000	0
SBE Conductivity PCB (included with sensors)	55385	1	32.6	0.0326	115.6	0.198	4.30	0.05135	0.000	0
Conductivity, PCB jumper, CA W26	52261	1	3.1	0.0031	112.8	0.170	3.80	0.04635	0.000	0
Shield, Radiation, C PCB + (4x) 2-56 x 1/4 BHCS	49887 / 55221	1	20.1	0.0201	115.6	0.198	4.80	0.05635	0.000	0
SBE Temperature PCB (included with sensors)	55388	1	22.0	0.022	115.6	0.198	-4.30	-0.03465	0.000	0
Temperature, PCB jumper, CA W25	52260	1	5.2	0.0052	112.8	0.170	1.30	0.02135	0.000	0
Shield, Radiation, T PCB + (4x) 2-56 x 1/4 BHCS	52371 / 55221	1	16.5	0.0165	115.6	0.198	-4.80	-0.03965	0.000	0
Oxygen, PCB jumper, CA W33 (only used with SBE 43f)	52265	1	6.1	0.0061	112.8	0.170	0.00	0.00835	0.000	0
Ribbon Cable (W13) (gray)	52248	1	29.6	0.0296	98.1	0.023	8.90	0.09735	0.000	0
Ribbon Cable (W14) (rainbow)	52249	1	37.1	0.0371	98.1	0.023	8.90	0.09735	0.000	0
CT Sensor Sail Assembly (stanchion, guard, nut, nacelle, (6x) 6-32 x 5/16 FHMS, o-ring (E70-016), o-ring (E70-012), SBE components)	52368	1	293.8	0.2938	121.6	0.258	17.90	0.18735	0.000	0
Mass Shifter Stability Rod Assy w/ Hard Stop ("the stinger")		1	87.6	0.0876	111.5	0.157	0.00	0.00835	0.000	0
Electronic rail, port	49846	1	60.9	0.0609	59.7	-0.361	0.00	0.00835	0.000	0
Electronics rail, starboard	52379	1	60.9	0.0609	59.7	-0.361	0.00	0.00835	0.000	0
Seaglider Mainboard (Mother board, TT8, CF8, flash card)	522990-B.4	1	263.5	0.2635	54.6	-0.412	0.70	0.01535	0.000	0
Insulation Sheet-OEM/Main	52370	1	4.0	0.004	55.2	-0.406	0.44	0.01275	0.000	0
(6x) 6-32 x 1/4 PHMS, main board to rails	55237	1	4.5	0.0045	54.6	-0.412	0.50	0.01335	0.000	0

Appendix A: Nominal

SEAGLIDER - TRIM AND BALANCE

SG 194

Date

24-Oct-13

Vehicle CG

x_CG

95.752

cm

z_CG

-0.835

cm

y_CG

0

cm

Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m
Assy, OEM Navigation Board (transponder PCB, transformer/inductor, capacitors, GPS reciever, W6 GPS serial interface, GPS antenna cable, DC block, Cable Assembly W7, all soldered on wiring) + RAFOS clock and receiver board & cable to mainboard		1	559.1	0.5591	54.6	-0.412	-0.43	0.00405	0.000	0
(6x) 6-32 x 1/4 PHMS, OEM board to rails	55237	1	4.5	0.0045	54.6	-0.412	-0.43	0.00405	0.000	0
Assy, Iridium Moedm and bracket (includes satellite reciever - iridium LBT, cable assy W3, all clamps for wires, RF switch, DC block, and mounting hardware)	52324	1	271.6	0.2716	54.9	-0.409	1.90	0.02735	0.000	0
4x) 6-32 x 1/4 PHMS + #6 LW, modem cage to rails	55209 / 55217	1	3.4	0.0034	52.8	-0.430	0.29	0.01125	0.000	0
Battery Pack, LV w/ CA W1	52237	1	2664.4	2.6644	54.9	-0.409	-1.81	-0.00975	0.000	0
Battery Pack Cage, LV	49847	1	71.6	0.0716	54.9	-0.409	-1.64	-0.00805	0.000	0
(6x) 6-32 x 1/4 PHMS + #6 LW, batt. cage to rails	55209 / 55217	1	5.4	0.0054	55.3	-0.405	-0.25	0.00585	0.000	0
Hull, Fwd Battery	49817	1	3646.0	3.646	110.6	0.148	0.00	0.00835	0.000	0
Hull, Aft Battery	49817	1	3638.6	3.6386	85.5	-0.103	0.00	0.00835	0.000	0
Joint Ring, battery hulls	49818	1	674.7	0.6747	98.9	0.031	0.00	0.00835	0.000	0
(8x) 6-32 x 3/4 SHCS + #6 LW, joint ring to fwd/aft batt. Hull	55214 / 55217	1	13.3	0.0133	98.1	0.023	0.00	0.00835	0.000	0
(2x) O-Ring, E70-271	52412	1	17.1	0.0171	98.1	0.023	0.00	0.00835	0.000	0
(12x) Clips, Rib	49883	1	25.4	0.0254	98.1	0.023	3.95	0.04785	0.000	0
(12x) Clips, Wiring	52405	1	18.7	0.0187	98.1	0.023	3.95	0.04785	0.000	0
Oil Absorption pad, 1.9" x 10.125" + (2x) rib clips	55527 / 49883	1	11.2	0.0112	98.1	0.023	3.95	0.04785	0.000	0
Endcap, Forward	49830	1	1897.1	1.8971	39.5	-0.563	0.00	0.00835	0.000	0
Acoustic transducer (ITC-3013), w/ cable W36	52267 / 52223	1	1438.7	1.4387	22.9	-0.729	0.00	0.00835	0.000	0
O-Ring, E70-238, transducer to Endcap	52413	1	3.2	0.0032	26.7	-0.691	0.00	0.00835	0.000	0
(6x) Alum 1/4-20 x 3/4 SHCS + 1/4 LW, xducer to end cap	55236 / 55252	1	15.2	0.0152	27.2	-0.686	0.00	0.00835	0.000	0
Hull, Electronics	49829	1	1696.0	1.696	58.0	-0.378	0.00	0.00835	0.000	0
O-Ring, E70-263, elec hull to fwd endcap	52410	1	7.1	0.0071	50.2	-0.456	0.00	0.00835	0.000	0
(8x) 6-32 x 1/2 SHCS + #6 LW, electric hull to fwd encap	55183 / 55217	1	10.3	0.0103	50.5	-0.453	0.00	0.00835	0.000	0
Compass bracket	49848	1	24.4	0.0244	30.5	-0.653	-0.35	0.00485	0.000	0
Compass w/compass carrier PCB	55305 / 52466	1	46.8	0.0468	30.5	-0.653	0.00	0.00835	0.000	0
(4x) 4-40 x 3/8 SHCS + #4 x 1/8L stand off	55256 / 55282	1	2.6	0.0026	30.5	-0.653	-0.31	0.00525	0.000	0
(2x) 4-40 x 3/8 SHCS + #4 LW, comp mt to fwd encap	55228 / 55233	1	1.3	0.0013	27.4	-0.684	-0.81	0.00025	0.000	0
Cable Assembly W9 (compass cable)	52244	1	21.5	0.0215	38.9	-0.569	0.38	0.01215	0.000	0
Total weight (SUM of components' weight col D)			44704	44.704	Excluding movable battery mass					
Total Scale Weight			58852							